

Example 1 : Basic Contradiction

$$\frac{\begin{array}{c} \frac{\frac{\frac{\neg p \wedge (q \wedge \neg q)^{[1]}}{q \wedge \neg q} [\wedge\text{-elim-2}]}{\neg q} [\wedge\text{-elim-1}]}{false} [\wedge\text{-elim-2}] \\ [contradiction] \\ false \\ [false\text{-elim}] \\ p \end{array}}{(\neg p \wedge (q \wedge \neg q)) \Rightarrow p} [\Rightarrow\text{-intro}^{[1]}]$$
$$\frac{\frac{\frac{\frac{p \wedge \neg p^{\neg[1]}}{[\wedge \text{-elim-1}]}}{p}}{[\wedge \text{-elim-2}]}}{\neg p}}{[\text{contradiction}]}}{\text{false}}{[\text{false-elim}]}}{q}}{(p \wedge \neg p) \Rightarrow q \quad [\Rightarrow \text{-intro}^{[1]}]}$$
$$\frac{\frac{\lceil p \wedge \neg p \rceil^{[1]}}{false} \text{ [contradiction]}}{p \wedge \neg p \Rightarrow false} [\Rightarrow \text{-intro}^{[1]}]$$
$$\frac{\frac{\frac{\overline{false}}{\neg p} \quad [\neg\text{-intro}^{[2]}]}{\neg(\neg p)} \quad [\Rightarrow\text{-elim}]}{\overline{\neg p \Rightarrow false} \Rightarrow \neg(\neg p)} \quad [\Rightarrow\text{-intro}^{[1]}]$$

1

Example 5 : Modus Tollens via Contradiction

Prove: $(p \Rightarrow q) \wedge \neg q \Rightarrow \neg p$

$$\frac{\frac{\neg q \quad [\Rightarrow \text{-elim}]}{q} \quad \frac{\text{false}}{\text{contradiction}} \quad [\text{assumption}]}{\frac{p}{\neg p} \quad [\neg \text{-intro}^{[2]}]} \quad [\Rightarrow \text{-intro}^{[1]}]$$

Classic modus tollens pattern using contradiction.

Example 6 : Disjunction from Contradiction

Prove: $\neg p \Rightarrow (p \Rightarrow q)$

$$\frac{\frac{\frac{\neg p}{p} \quad [\neg \text{-intro}^{[2]}]}{\text{false}} \quad [\text{contradiction}]}{\frac{q}{(p \Rightarrow q)} \quad [\Rightarrow \text{-intro}^{[2]}]} \quad [\Rightarrow \text{-intro}^{[1]}]$$

From $\neg p$, we can prove $p \Rightarrow q$ for any q.

Example 7 : Double Negation Elimination

Prove: $\neg \neg p \Rightarrow p$ (*requires(classical)(logic)*)

$$\frac{\frac{\frac{\neg \neg p}{\neg p} \quad [\neg \text{-intro}^{[2]}]}{\text{false}} \quad [\text{contradiction}]}{\frac{p}{\neg (\neg (\neg p))} \quad [\neg \text{-intro}^{[2]}]} \quad [\Rightarrow \text{-intro}^{[1]}]$$

In classical logic, not $\neg p$ implies p. This requires excluded middle or equivalent axiom.

Example 8 : Reductio ad Absurdum

Prove: $(\neg p \Rightarrow (q \wedge \neg q)) \Rightarrow p$

$$\frac{\frac{\frac{\frac{\neg \neg p}{\neg p} \quad [\neg \text{-intro}^{[2]}]}{q \wedge \neg q} \quad [\Rightarrow \text{-elim}]}{\frac{q}{\neg q} \quad [\wedge \text{-elim-1}]} \quad [\wedge \text{-elim-2}]} \quad [\text{contradiction}] \quad [\text{false-elim}]}{\frac{p}{\neg (\neg p)} \quad [\neg \text{-intro}^{[2]}]} \quad [\Rightarrow \text{-intro}^{[1]}]$$

Reductio ad absurdum: if assuming not p leads to absurdity, then p holds.

Example 9 : Contradiction with Universal Quantifier

Prove: $(\forall x \bullet \mathbb{P} x) \wedge \exists x \bullet \neg (\mathbb{P} x)$ is contradictory.

$$\frac{\frac{\frac{\frac{\frac{\frac{\frac{\vdash (\forall x \bullet \mathbb{P} x) \wedge (\exists x \bullet \neg (\mathbb{P} x))}{\vdash \neg (\mathbb{P} a)} [\text{contradiction}]}}{\vdash \mathbb{P} a} [\text{fresh } a]}{\vdash \neg (\mathbb{P} a)} [\text{fresh } a]}]{\vdash \exists x \bullet \neg (\mathbb{P} x)} [\exists \text{ elim}]}]{\vdash \forall x \bullet \mathbb{P} x} [\forall \text{-elim}]}]{\vdash (\forall x \bullet \mathbb{P} x) \wedge (\exists x \bullet \neg (\mathbb{P} x))} [\wedge \text{-elim-1}]$$

If something holds for all x , it cannot fail for some x .

Example 10 : Proving Uniqueness by Contradiction

Prove: if f is injective, then $f(x) = f(y) \Rightarrow x = y$.

$$\frac{\frac{\frac{\top x \neq y \neg^{[2]}}{f(x) \neq f(y)} \text{ [injective property]}}{\text{false}} \text{ [contradiction with } f(x) = f(y)]}{x = y} \text{ } \neg\text{-intro}^{[2]}$$

$$\frac{}{(injective(f) \wedge f(x) = f(y)) \Rightarrow x = y} \text{ } \Rightarrow\text{-intro}^{[1]}$$

Uses contradiction to prove equality.

Example 11 : Case Analysis Leading to Contradiction

Prove: $p \vee q, \neg p, \neg q \Rightarrow false$

$$\frac{\frac{\overline{false} \text{ [contradiction with } \neg p]}{\overline{false}} \quad \frac{\overline{false} \text{ [contradiction with } \neg q]}{\overline{false} \text{ [\ve elim from } p \vee q]}}{\overline{false} \text{ [\Rightarrow -intro}^{[1]}]}} \frac{}{((p \vee q) \wedge \neg p \wedge \neg q) \Rightarrow false}$$

Both cases lead to contradiction, so the premises are inconsistent.

Example 12 : Best Practices for Contradiction Proofs

When using proof by contradiction: 1. Clearly mark the assumption you're contradicting with [assumption N] 2. Show explicitly where false is derived 3. Use [not intro from N] to discharge the assumption 4. Document the contradiction (what conflicts with what) 5. In natural deduction, false elim lets you conclude anything 6. Remember: contradiction is a classical technique (not constructive)